

SAT Hard Question Set 4

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Novel Prep

Overview

Question 1

The incomplete table below summarizes the number of left-handed and right-handed students by gender for eighth-grade students at Keisel Middle School.

Gender	Left	Right
Female		
Male		
Total	18	122

There are **5 times** as many right-handed female students as left-handed female students, and **9 times** as many right-handed male students as left-handed male students.

If there are a total of 18 left-handed and 122 right-handed students, which of the following is closest to the probability that a randomly selected right-handed student is female?

- A. 0.410
- B. 0.357
- C. 0.333

Answer 1

Correct Answer: A

Question 2

If shoppers enter a store at an average rate of r shoppers per minute and each stays in the store for an average time of T minutes, then the average number of shoppers in the store, N , is given by the formula:

$$N = rT$$

The owner of the Good Deals Store estimates that 3 shoppers per minute enter the store, and each stays for 15 minutes. The store owner estimates 45 shoppers in the store at a time.

Now, for a new store, the estimate is 90 shoppers per hour, each staying 12 minutes. What percent less is this new store's average than the original?
(Note: Ignore the percent symbol in your answer.)

- A. 57
- B. 58
- C. 59
- D. 60

Answer 2

Correct Answer: D

Question 3

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week.

For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3 hours per week**.

Hours practiced	6 to 13 points	14 or more points	Total
3 to 5 hours	6	4	10
More than 5 hours	4	26	30
Total	10	30	40

Which of the following is the **largest population** to which the results of the study can be generalized?

- A. All trivia teams in the tournament that scored 14 or more points.
- B. The 55 trivia teams in the sample.
- C. The 40 trivia teams in the sample that practiced 3+ hours/week.
- D. All trivia teams in the tournament.

Answer 3

Correct Answer: D

Question 4

A circle is centered at $(-1, 1)$. Line t is tangent to the circle at point $(4, -3)$. Which of the following points lies on line t ?

A. $\left(0, \frac{5}{4}\right)$

B. $(3, 6)$

C. $(8, 2)$

D. $(9, 1)$

Answer 4

Correct Answer: C

Question 5

Two identical rectangular prisms have a height of 90 cm, square bases, and surface area $K \text{ cm}^2$. Glued along a square base, the resulting surface area is $\frac{92}{47}K \text{ cm}^2$.

What is the side length s , in cm, of the square base?

Answer 5

Final Answer: 8

Question 6

In triangle RST , angle T is a right angle. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK} \parallel \overline{RT}$.

Given:

- $RT = 72$ units
- $LK = 24$ units
- Area of $\triangle RST = 792$ square units

Find the length of $\overline{KT}$, in units.

Answer 6

Final Answer: $\boxed{\frac{22}{3}}$

Question 7

The equation of a circle is:

$$(x - 6)^2 + (y + 5)^2 = 16$$

and point $P(10, -5)$ lies on the circle. If $\overline{PQ}$ is the diameter, what are the coordinates of point Q ?

Answer 7

Final Answer: $2, -5$

Thank You!



We appreciate your time and focus.

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